

Roshan Hegde

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EDUCATION

University of Illinois at Urbana-Champaign

May 2027

B.S. in Computer Science, Grainger College of Engineering

GPA: 3.80/4.0

B.S. in Mathematics, College of Liberal Arts & Sciences

RELEVANT COURSEWORK

Courses: Advanced Algorithms, Operating Systems, Programming Languages and Compilers, Abstract Algebra, Linear Algebra, Graph Theory, Cryptography, Data Structures, Probability and Statistics

SKILLS

Languages: C, C++, Rust, Go, Java, Kotlin, Python, Typescript

Tools: Git, Docker, Linux, Nix, Vim

EXPERIENCE

Grainger | *SWE Intern* | *Typescript, PostgreSQL, AWS*

June - August 2026

- Contributed directly to the full-stack application Grainger Workstation, serving 1000+ users
- Built a code migration tool used across 12 teams to move off Mocha onto Vitest, reducing CI/CD times by 80%
- Hybridized AST transformations with agentic reasoning to minimize manual work without sacrificing correctness

OFI, Chicago IL | *SWE Intern* | *Azure, python*

May – August 2025

- Analyzed current utilization of cloud assets and identified opportunities to improve cost efficiency by 12%
- Developed a data analysis service to automate VM performance reporting to identify over-allocated resources
- Proposed a plan to improve resource cost-effectiveness to the infrastructure team

University Of Illinois Urbana-Champaign | *Operating Systems Course Assistant* | *C, GDB* August 2025 - Present

- Developed and reviewed programming assignments for students taking the course
- Answered both conceptual and implementation questions while hosting office hours

PROJECTS

MechMania, Champaign IL | *Lead Developer* | *Go, Rust, Python, Nix*

August 2024 – Present

- Built a multithreaded web-socket server in Go interacting with AWS services including databases and cloud storage
- Reworked a web-socket IPC protocol to shared memory, resulting in a >1000x speedup
- Hardened security by utilizing Nix and bubble-wrap to run arbitrary code without vulnerabilities

LumonOS: Operating System | *C, Unix, RISC-V, QEMU, GDB*

January – August 2025

- Created fully functional, RISC-V, Unix-based pre-emptive operating system with round robin scheduling policy.
- Built a virtio block driver, cryptography driver, and entropy driver
- Implemented support for pipes between processes and argument passing according to POSIX standard
- Placed 3rd in the Operating System design competition hosted by the ECE department

AWARDS

1st Place, GPT2 Inference Optimization Competition | *Cuda, C, PTX, Nsight Compute*

December 2025

- Created custom CUDA kernels, profiled with NVIDIA Nsight Compute, achieving a 120x speedup over the baseline and 2x speedup over NVIDIA cuBLAS
- Implemented and tuned optimizations including flash attention, tensor cores, shared memory tiling, and fusion

1st place, Hackillinois '24, '25 | *Rust, Cargo*

February 2024 and 2025

- Competed with over 1,100 other students at Illinois' largest Hackathon, winning the General and DevTool tracks

1st place, MechMania '24, '25 | *Java, Python*

September 2023 and 2024

- Fine-tuned heuristics to control human and zombie swarms and to pilot dogfighting planes